

Graphics Hardware

Computer Graphics

Interactive input devices

There are various input devices with computer among them

Digitizer(Graphics tablets), light pen and **Touch panels** are interactive input devices.

A digitizer is a hardware device that receives analog information, such as sound or light, and records it digitally. Usually, the information is stored in a file on a computing device. This process is called digitization.

Digitizer examples

For example, a digital camera is a digitizer. Light enters through the camera lens, and the hardware and software inside the camera converts that information to binary data, and stores it an image file. The user may then transfer the file to a computer, where he or she can edit the image, print it out.

Digitizer examples

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Camera

Digital tablets

Audio digitizer

Scanner

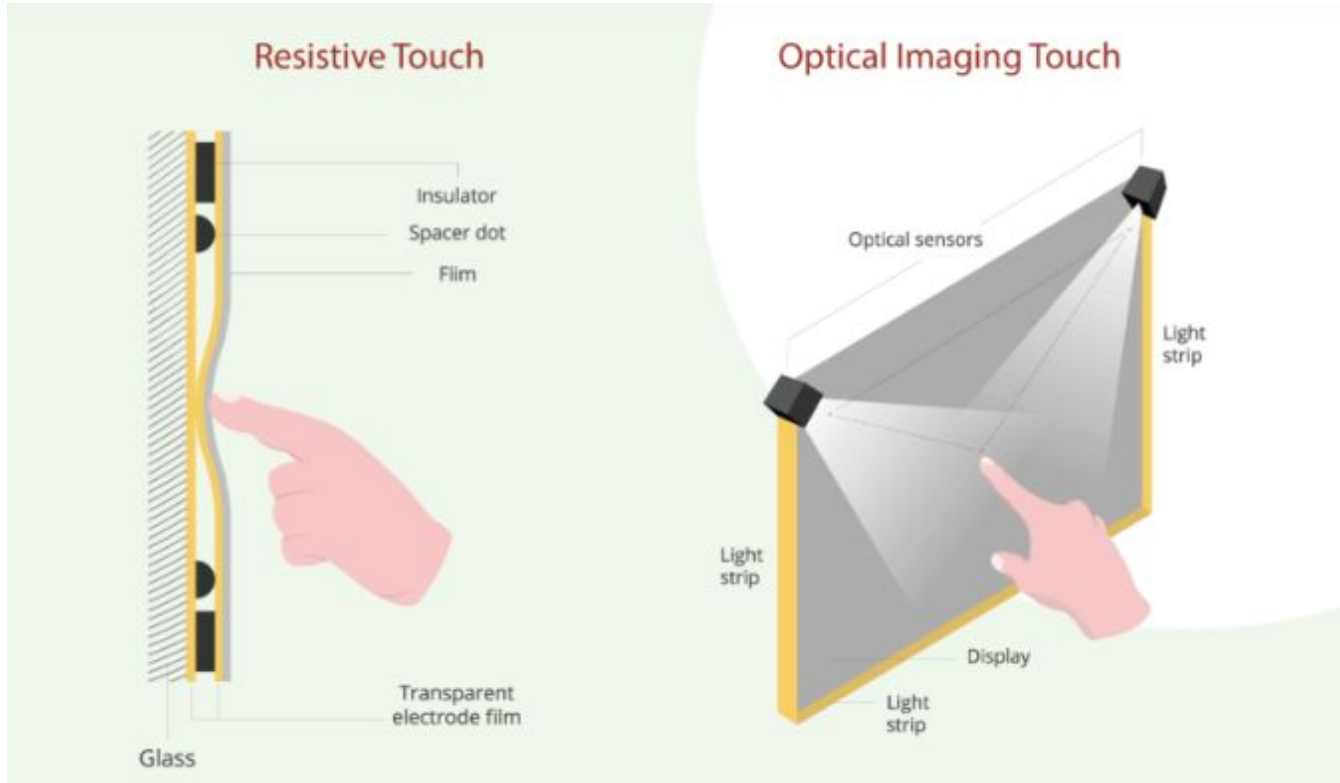
Touch Pannels

Touch Panels is a type of display screen that has a touch-sensitive transparent panel covering the screen. A touch screen registers input when a finger or other object comes in contact with the screen.

When the wave signals are interrupted by some contact with the screen, that located is recorded.

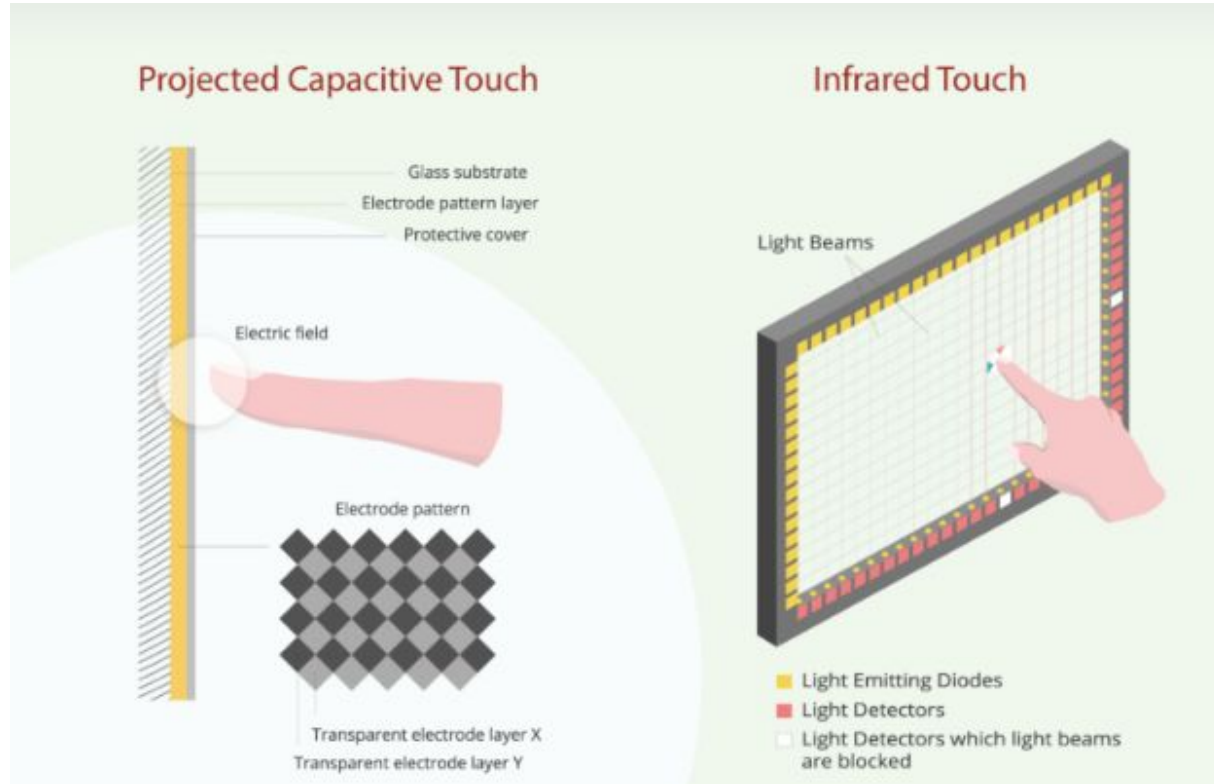
Types of Touch Pannels

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Types of Touch Pannels

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Light Pen

Light Pen (similar to the pen) is a pointing device which is used to select a displayed menu item or draw pictures on the monitor screen.

It consists of a photocell and an optical system placed in a small tube.

When its tip is moved over the monitor screen, and pen button is pressed, its photocell sensing element detects the screen location and sends the corresponding signals to the CPU.

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Display Device and Hardcopy Devices

Introduction

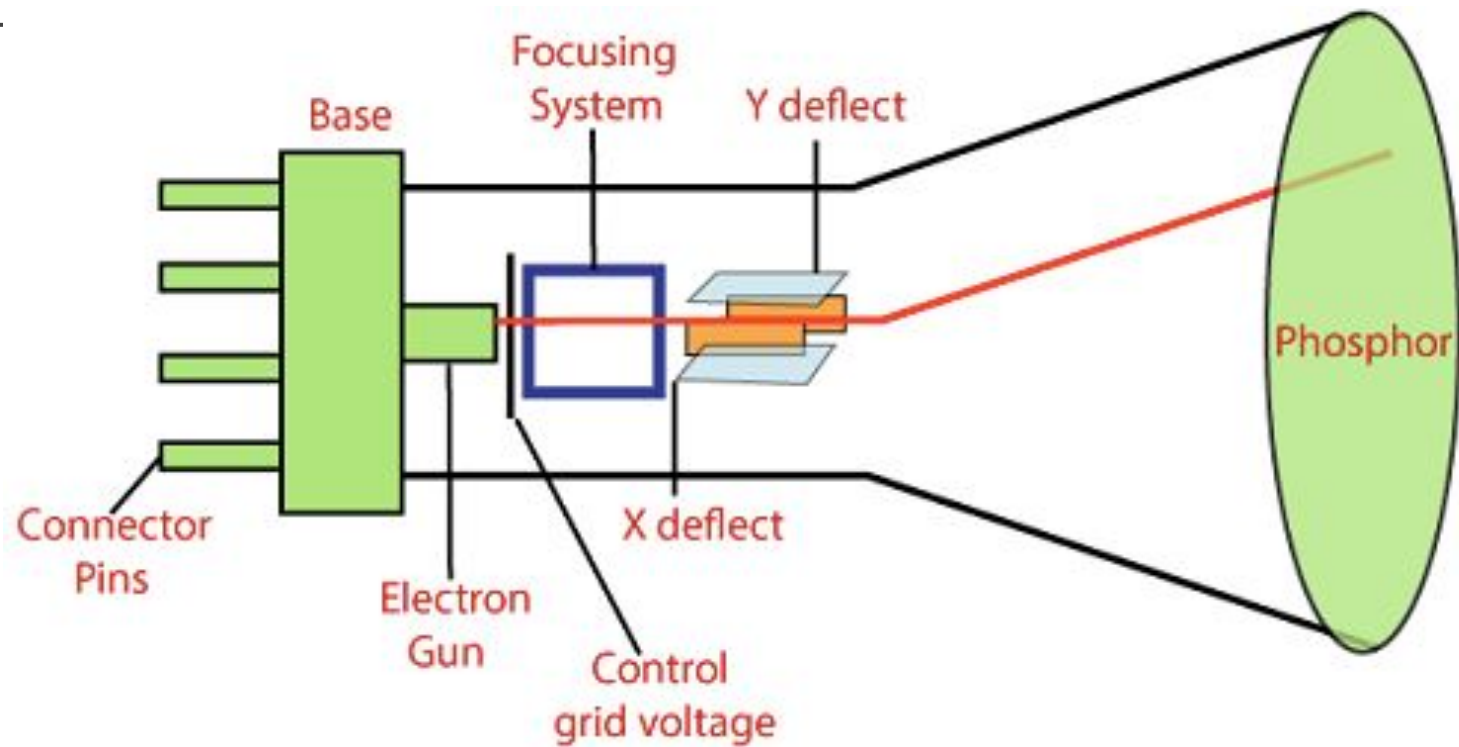
The display device is an output device used to represent the information in the form of images. Display systems are mostly called a video monitor or Video display unit.

- Cathode-Ray Tube(CRT)
- Color CRT Monitor
- Liquid crystal display(LCD)
- Light Emitting Diode(LED)
- Direct View Storage Tubes
- Plasma Display

CRT

- CRT stands for Cathode ray tube. It is a technology which is used in traditional computer monitor and television.
- Cathode ray tube is a particular type of vacuum tube that displays images when an electron beam collides on the radiant surface.

CRT



Components of CRT

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- **Electron Gun:** The electron gun is made up of several elements, mainly a heating filament and a cathode. The electron gun is a source of electrons focused on a narrow beam facing the CRT.
- **Focusing & Accelerating Anodes:** These anodes are used to produce a narrow and sharply focused beam of electrons.
- **Horizontal & Vertical Deflection Plates:** These plates are used to guide the path of the electron the beam. The plates produce an electromagnetic field that bends the electron beam through the area as it travels.
- **Phosphorus-coated Screen:** The phosphorus coated screen is used to produce bright spots when the high-velocity electron beam hits it.

Color CRT monitor

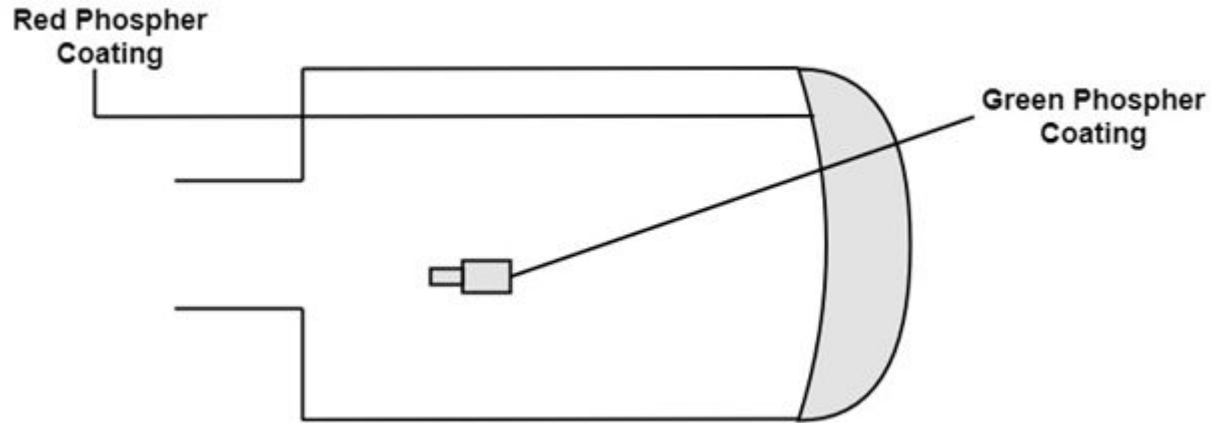
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The CRT Monitor display by using a combination of phosphors. The phosphors are different colors. There are two popular approaches for producing color displays with a CRT are:

- Beam Penetration Method
- Shadow-Mask Method

Beam Penetration Method

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Beam Penetration Method

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- This method is commonly used in random scan system.
- In this method, the CRT screen is coated with two layers of phosphor, red and green and the displayed color depends on how far the electron beam penetrates the phosphor layers.
- This method produces four colors only, red, green, orange and yellow.
- A beam of slow electrons excites the outer red layer only; hence screen shows red color only.
- A beam of high-speed electrons excites the inner green layer. Thus screen shows a green color.

Beam Penetration Method

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Advantages:

- Inexpensive

Disadvantages:

- Only four colors are possible
- Quality of pictures is not as good as with another method.

Shadow Mask Method

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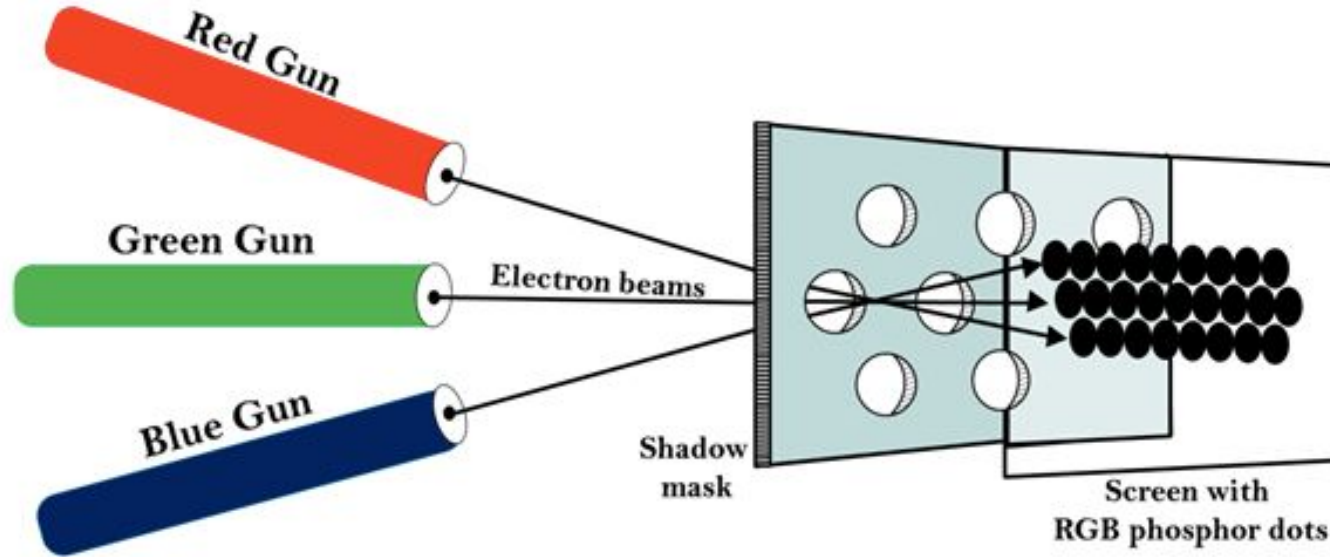
Shadow Mask Method is commonly used in Raster-Scan System because they produce a much wider range of colors than the beam-penetration method.

It is used in the majority of color TV sets and monitors. A shadow mask CRT has 3 phosphor color dots at each pixel position.

- One phosphor dot emits: red light
- Another emits: green light
- Third emits: blue light

This type of CRT has 3 electron guns, one for each color dot and a shadow mask grid just behind the phosphor coated screen.

Beam Penetration Method



Shadow Mask Method

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Advantage:

- Realistic image
- Million different colors to be generated
- Shadow scenes are possible

Disadvantage:

- Relatively expensive compared with the monochrome CRT.
- Relatively poor resolution
- Convergence Problem

Shadow Mask Method Working Mechanism

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The deflection system of the CRT operates on all 3 electron beams simultaneously; the 3 electron beams are deflected and focused as a group onto the shadow mask, which contains a sequence of holes aligned with the phosphor-dot patterns.

When the three beams pass through a hole in the shadow mask, they activate a dotted triangle, which occurs as a small color spot on the screen.

The phosphor dots in the triangles are organized so that each electron beam can activate only its corresponding color dot when it passes through the shadow mask.

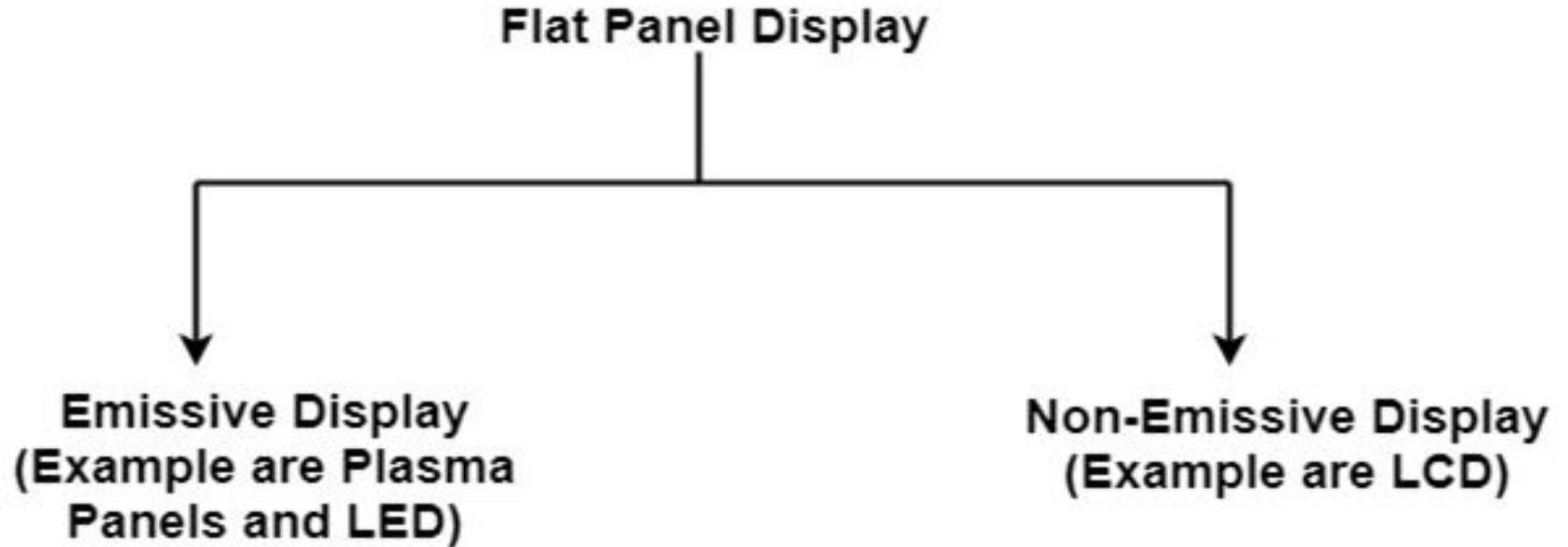
Flat Panel Display

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The Flat-Panel display refers to a class of video devices that have reduced volume, weight and power requirement compare to CRT.

Example: Small T.V. monitor, calculator, pocket video games, laptop computers, an advertisement board in elevator.

Flat Panel Display



Flat Panel Display

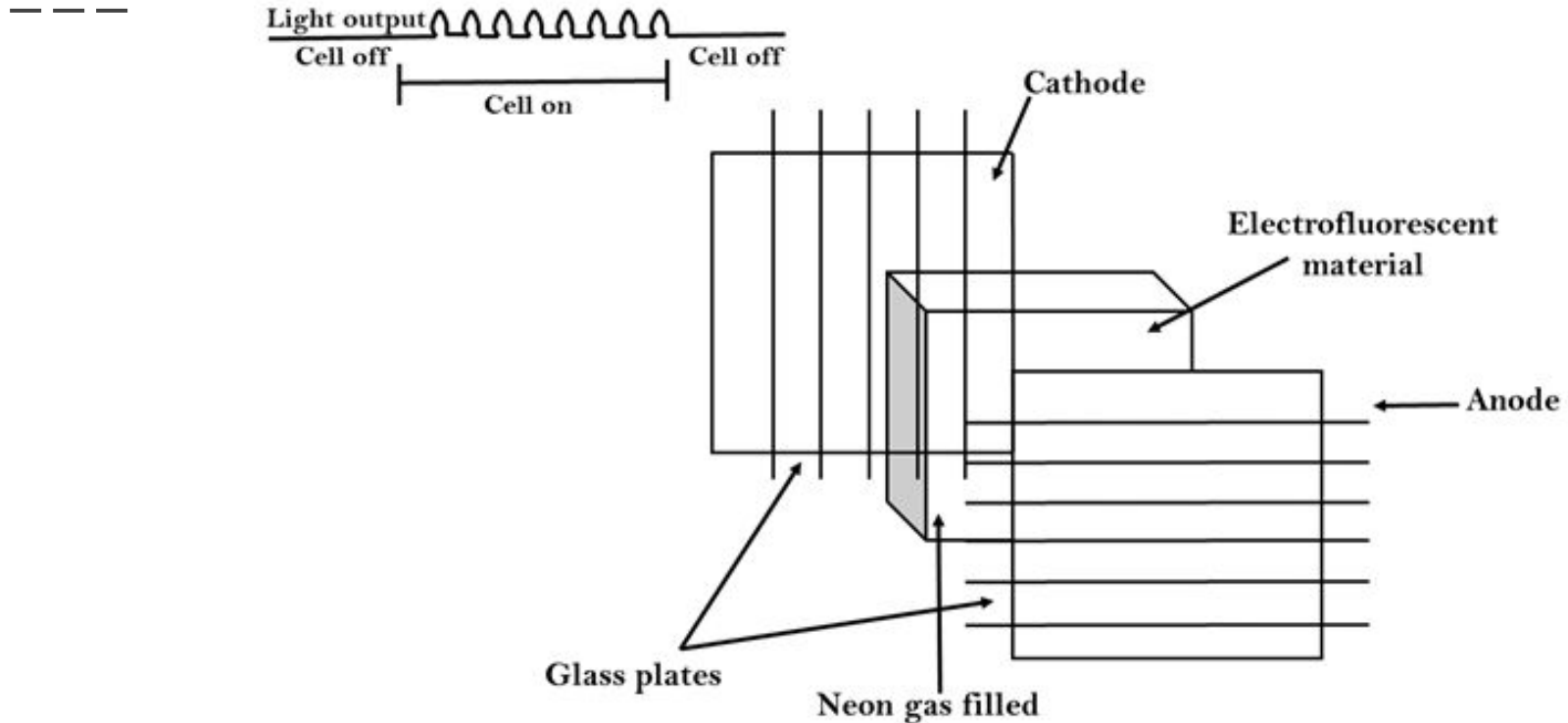
1. Emissive Display: The emissive displays are devices that convert electrical energy into light. Examples are Plasma Panel, thin film electroluminescent display and LED (Light Emitting Diodes).
2. Non-Emissive Display: The Non-Emissive displays use optical effects to convert sunlight or light from some other source into graphics patterns. Examples are LCD (Liquid Crystal Device).

Plasma Panel Display

Plasma-Panels are also called as Gas-Discharge Display. It consists of an array of small lights. Lights are fluorescent in nature. The essential components of the plasma-panel display are:

- **Cathode:** It consists of fine wires. It delivers negative voltage to gas cells. The voltage is released along with the negative axis.
- **Anode:** It also consists of line wires. It delivers positive voltage. The voltage is supplied along positive axis.
- **Fluorescent cells:** It consists of small pockets of gas liquids when the voltage is applied to this liquid (neon gas) it emits light.
- **Glass Plates:** These plates act as capacitors. The voltage will be applied, the cell will glow continuously.

Plasma Panel Display



Plasma Panel Display

Each cell of plasma has two states, so cell is said to be stable. Displayable point in plasma panel is made by the crossing of the horizontal and vertical grid. The resolution of the plasma panel can be up to $512 * 512$ pixels.

Advantage:

- High Resolution
- Large screen size is also possible.
- Less Volume
- Less weight
- Flicker Free Display

Plasma Panel Display

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Disadvantage:

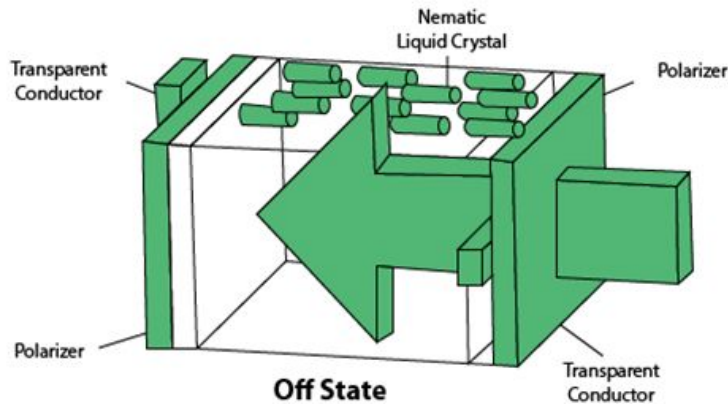
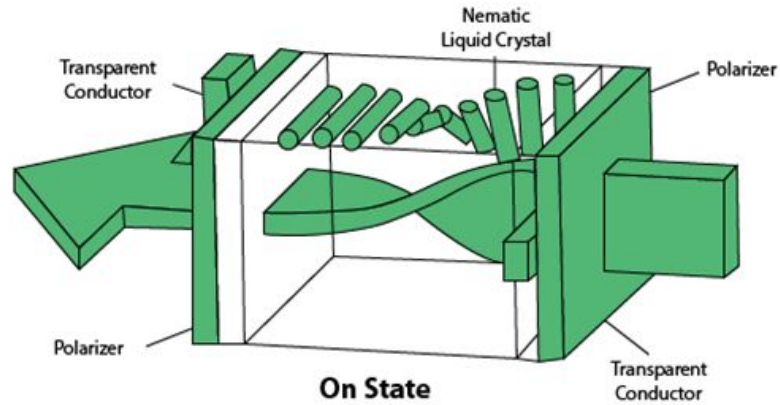
- Wiring requirement anode and the cathode is complex.
- Its addressing is also complex.

LED (Light Emitting Diode)

In an LED, a matrix of diodes is organized to form the pixel positions in the display and picture definition is stored in a refresh buffer.

Data is read from the refresh buffer and converted to voltage levels that are applied to the diodes to produce the light pattern in the display.

LCD (Liquid Crystal Display)



LCD (Liquid Crystal Display)

Liquid Crystal Displays are the devices that produce a picture by passing polarized light from the surroundings or from an internal light source through a liquid-crystal material that transmits the light.

LCD uses the liquid-crystal material between two glass plates; each plate is the right angle to each other between plates liquid is filled. One glass plate consists of rows of conductors arranged in vertical direction. Another glass plate is consisting of a row of conductors arranged in horizontal direction. The pixel position is determined by the intersection of the vertical & horizontal conductor. This position is an active part of the screen.

Liquid crystal display is temperature dependent. It is between zero to seventy degree Celsius. It is flat and requires very little power to operate.

LCD (Liquid Crystal Display)

Advantage:

- Low power consumption.
- Small Size
- Low Cost

Disadvantage:

- LCDs are temperature-dependent ($0-70^{\circ}\text{C}$)
- LCDs do not emit light; as a result, the image has very little contrast.
- LCDs have no color capability.
- The resolution is not as good as that of a CRT.

Hardcopy Devices

All the output devices can be categorised into two categories

- Hard Copy Devices
- Soft Copy Devices

Hard copy devices are those that give the output in the tangible form.

Printers and **Plotters** are two common hard copy devices.

Soft copy devices give output in the intangible form or the virtual form, e.g. something displayed on a screen. All the computer monitors are covered under this category.

Plotters

- Plotter is a printing device which can draw continuous lines. This is useful to print vector graphics rather than raster graphics unlike normal printers. Plotters are widely used in applications like CAD.
- Plotters print by moving one or more pen across the surface of a piece of paper. This means that plotters are restricted to line art, rather than raster graphics as with other printers.
- Pen plotters can draw complex line art, including text, but do so slowly because of the mechanical movement of the pens. They are often incapable of efficiently creating a solid region of colour, but can draw an area by drawing a number of close, regular lines.
- Plotters offered the fastest way to efficiently produce very large drawings or colour high-resolution vector-based artwork when computer memory was very expensive and processor power was very limited.

Plotters

There are a number of different types of plotters:

- **A drum plotter** draws on paper wrapped around a drum which turns to produce one direction of the plot, while the pens move to provide the other direction.
- **A flatbed plotter** draws on paper placed on a flat surface; and an electrostatic plotter draws on negatively charged paper with positively charged toner.

Pen plotters have essentially become obsolete, and have been replaced by large-format inkjet printers and toner based printers.

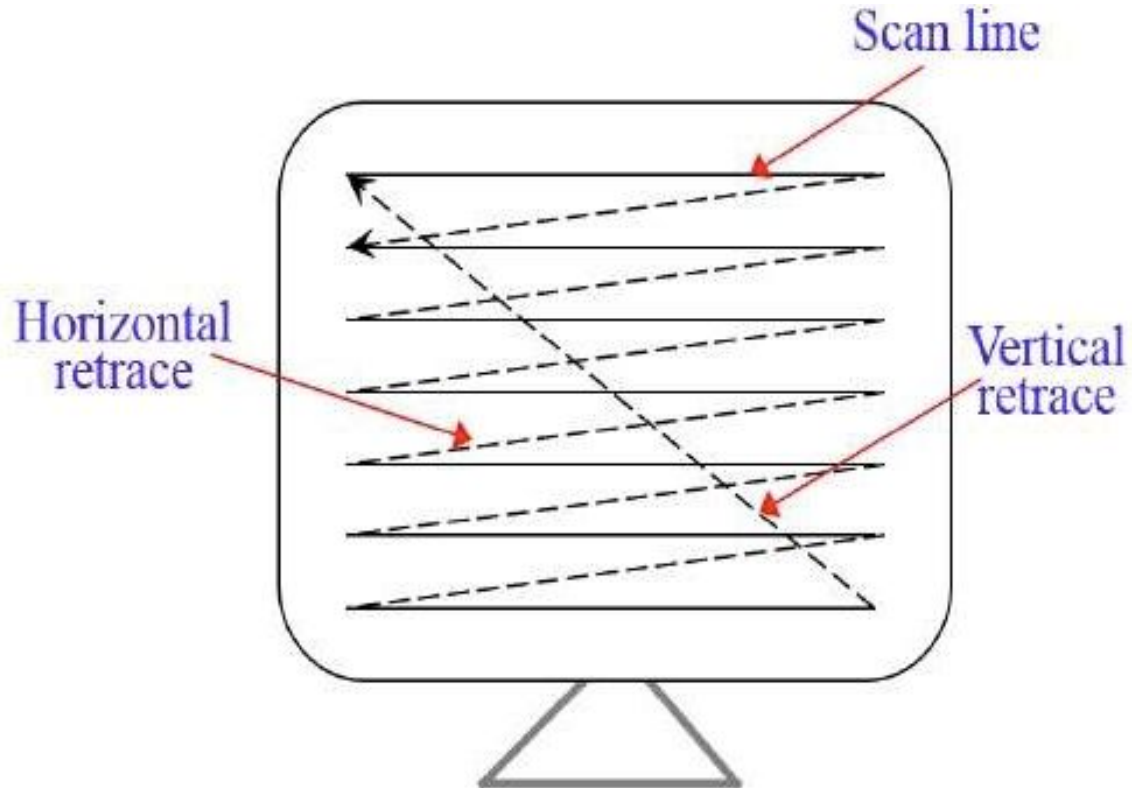
They are most frequently used for **CAE (computer-aided engineering)** applications, such as **CAD (computer-aided design)** and **CAM (computer-aided manufacturing)**.

Raster Scan Display

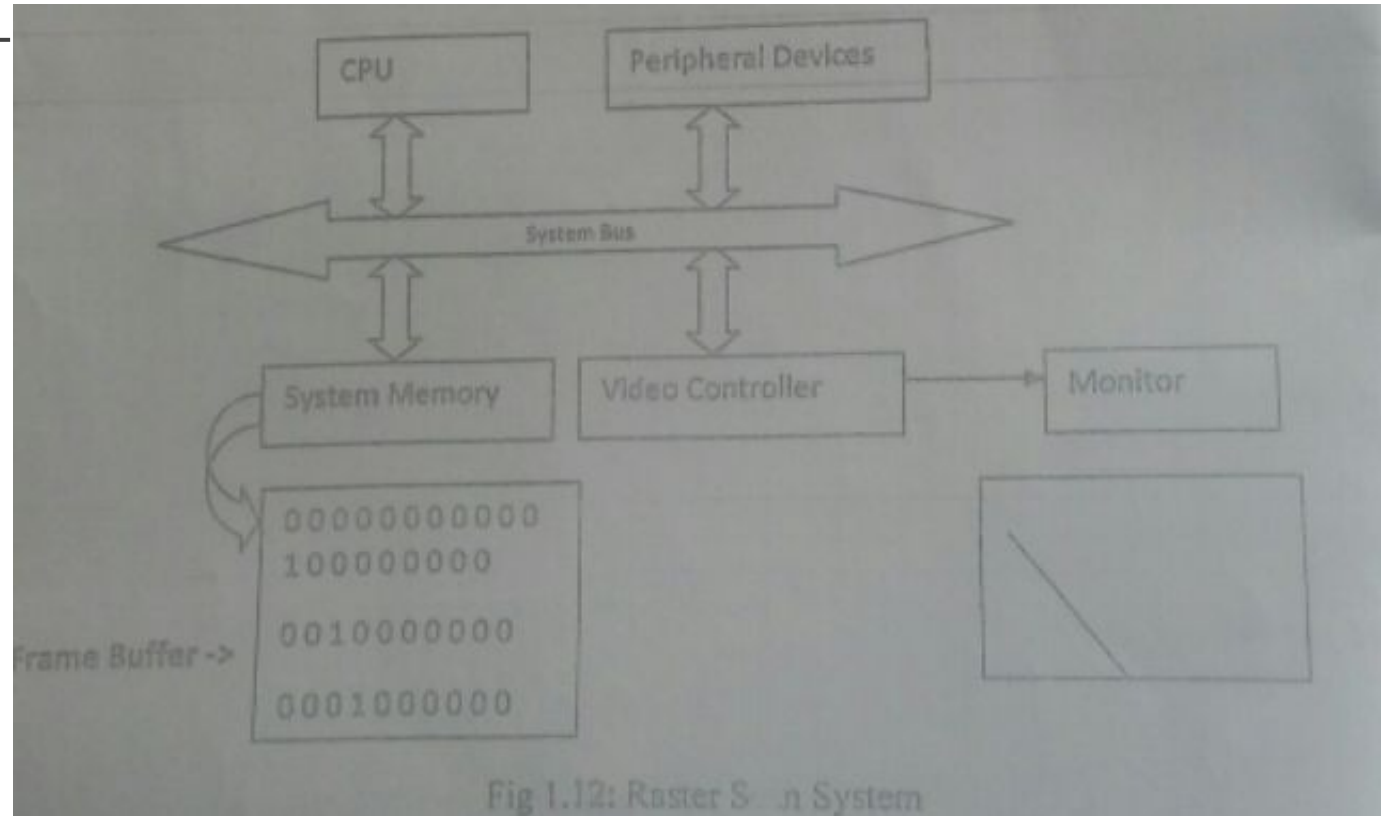
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- In a raster scan system, the **electron beam is swept across the screen, one row at a time from top to bottom**. As the electron beam moves across each row, the beam intensity is turned on and off to create a pattern of illuminated spots.
- Picture definition is stored in memory area called the **Refresh Buffer or Frame Buffer**. This memory area holds the set of intensity values for all the screen points. Stored intensity values are then retrieved from the refresh buffer and **“painted”** on the screen one row

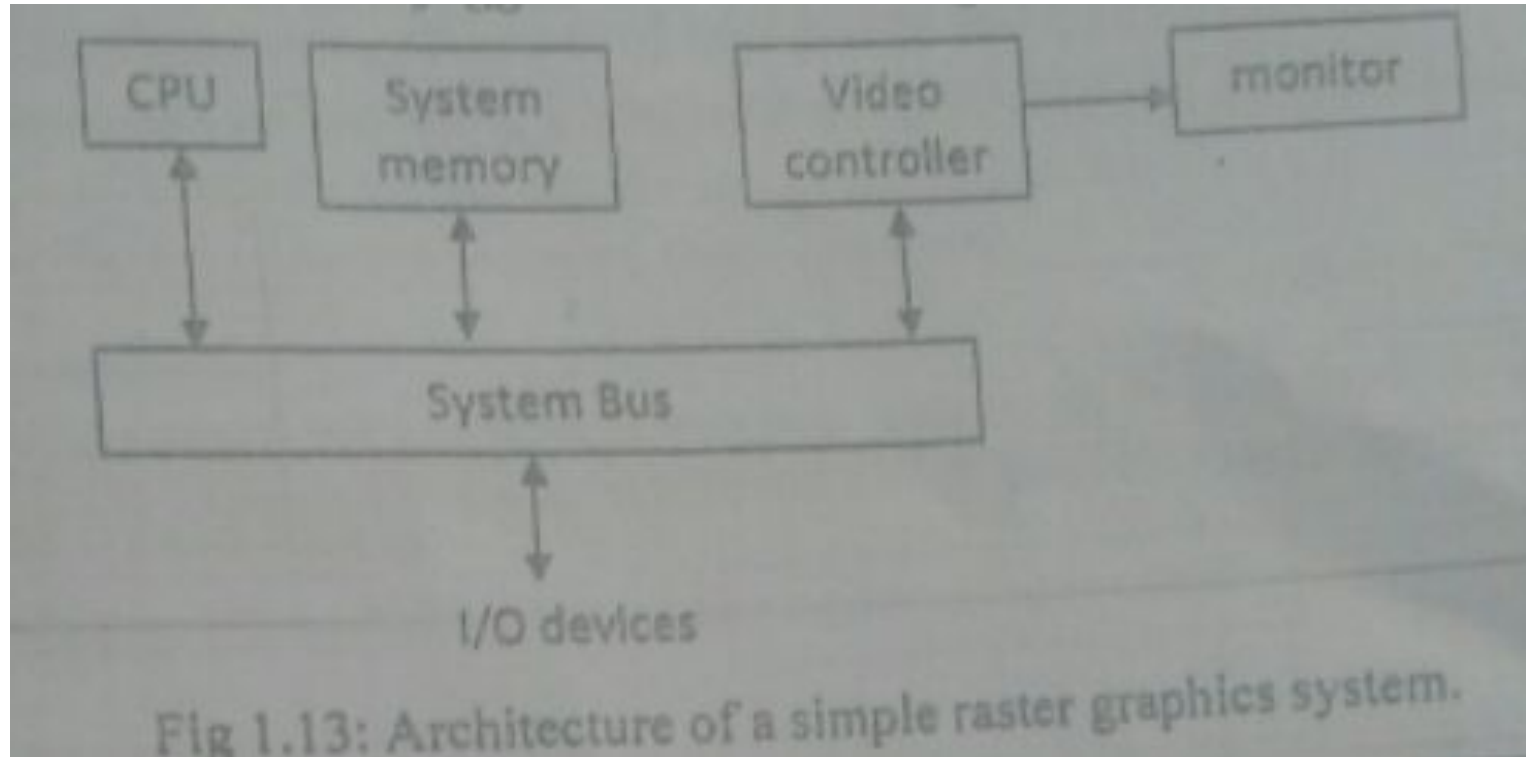
Raster Scan Display



Simple Raster Scan System



Architecture of Simple Raster scan system



Architecture of Raster scan system with frame buffer

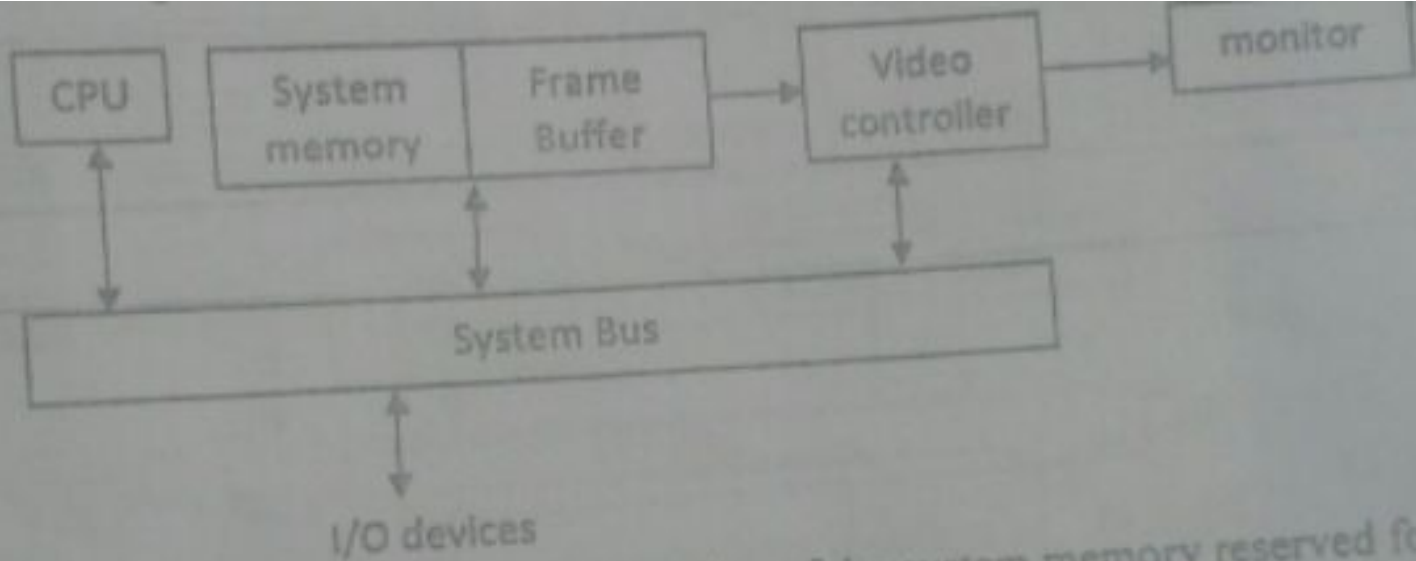
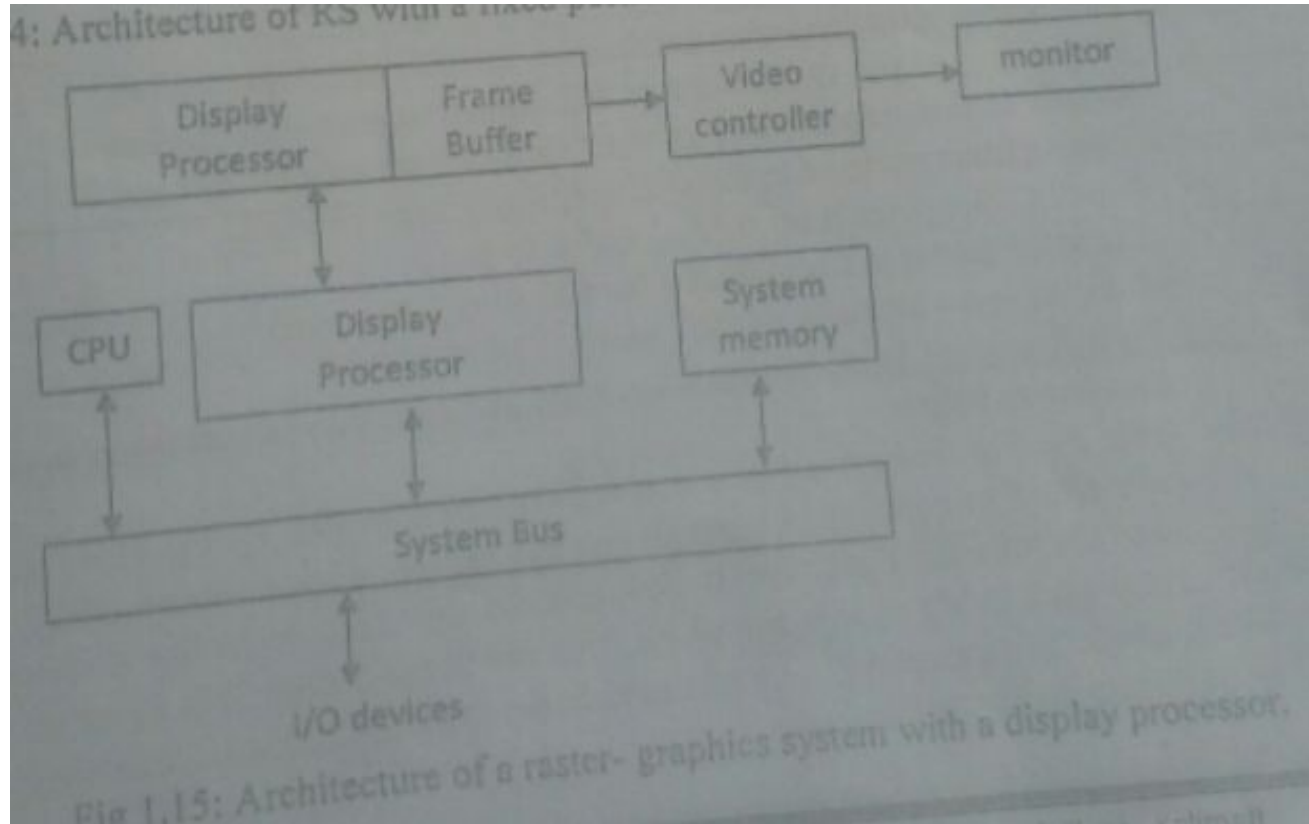


Figure 1.4: Architecture of RS with a fixed portion of the system memory reserved for the FB.

Architecture of Raster scan system With display processor



Advantages and disadvantages of RS

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- Advantages
 - It has ability to fill the areas with solid color or patterns
 - Time required for refreshing is independent of the complexity of image.
 - Low cost
- Disadvantages
 - Due to scan conversion Jaggies and star-casing

Jaggies

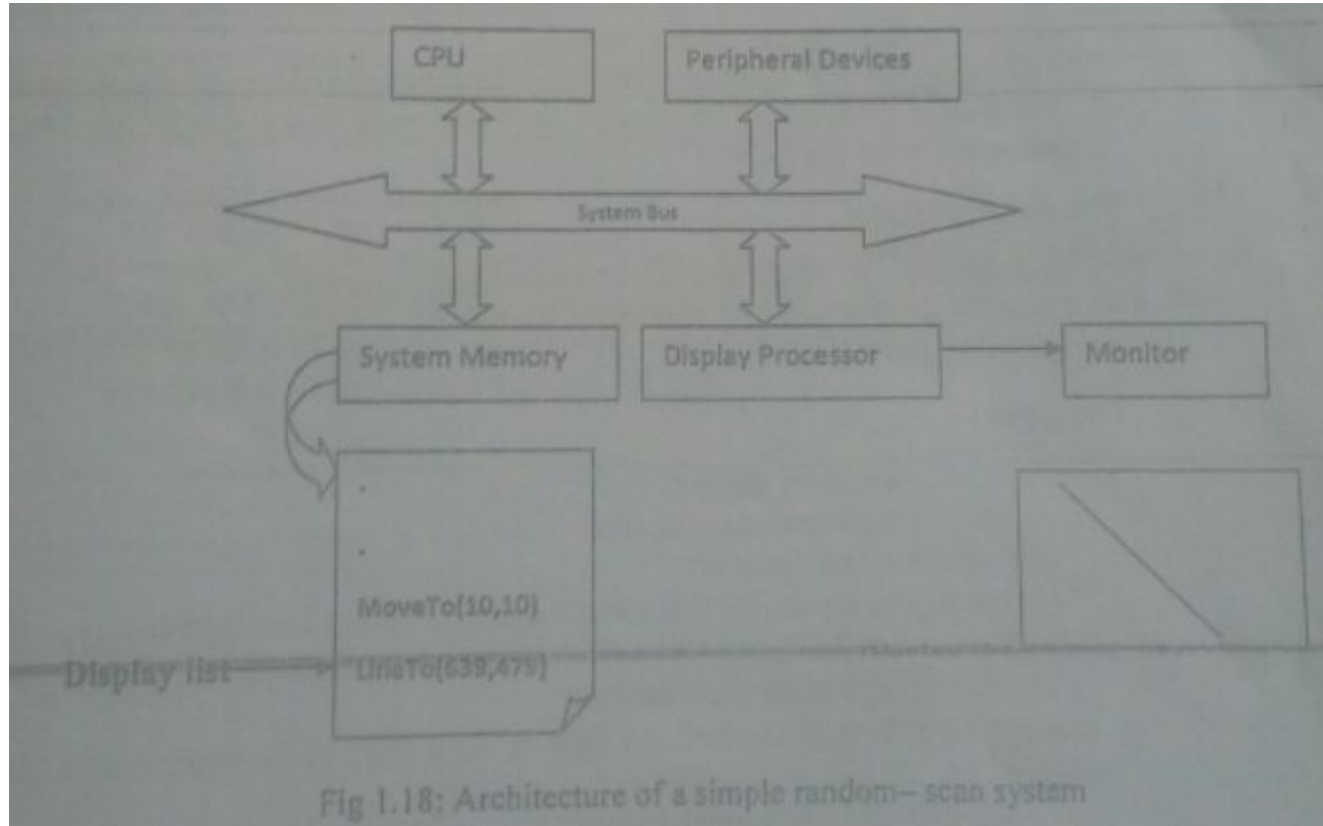


Random Scan Display

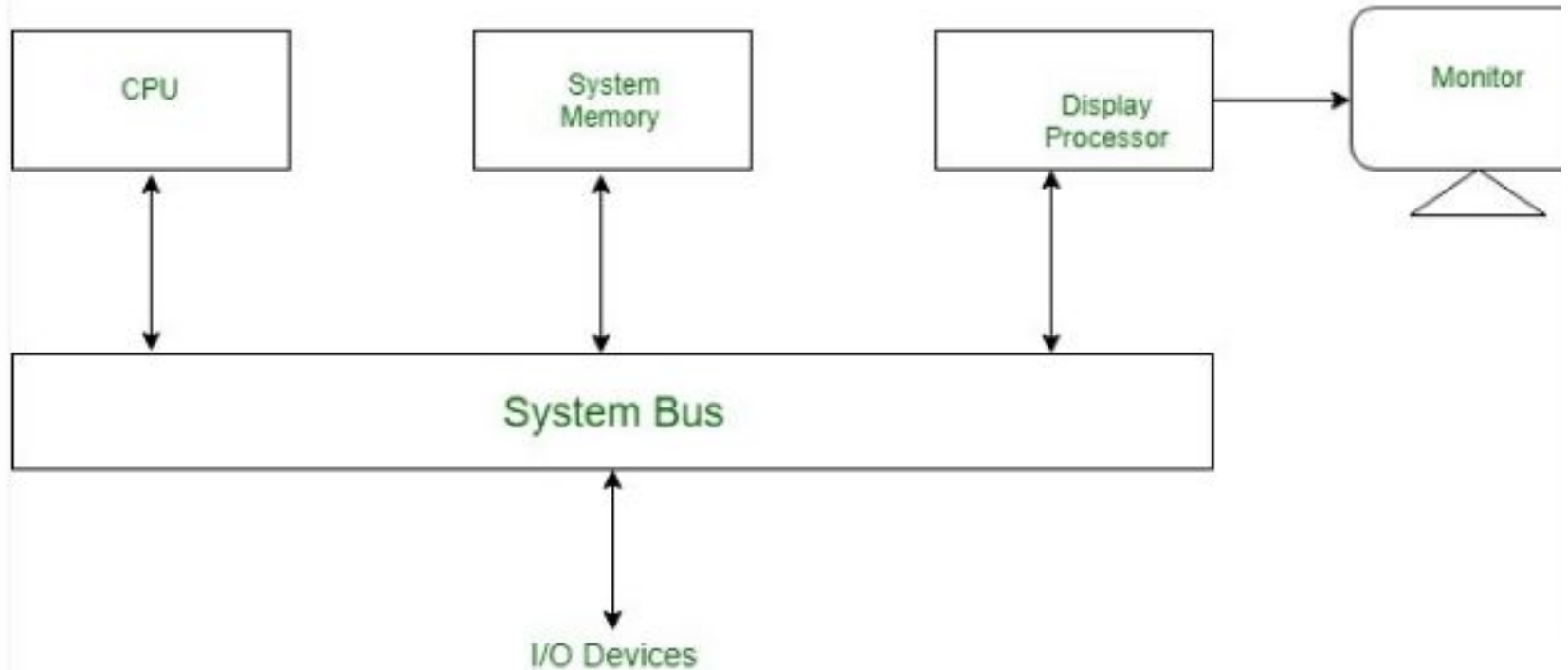
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- In this technique, the electron beam is directed only to the part of the screen where the picture is to be drawn rather than scanning from left to right and top to bottom as in raster scan. It is also called vector display, stroke-writing display, or calligraphic display.
- Picture definition is stored as a set of line-drawing commands in an area of memory referred to as the refresh display file.

Simple Random Scan System



Simple Random Scan System



Random Scan Display

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ADVANTAGES:

- Higher resolution as compared to raster scan display.
- Produces smooth line drawing.
- Less Memory required.

DISADVANTAGES:

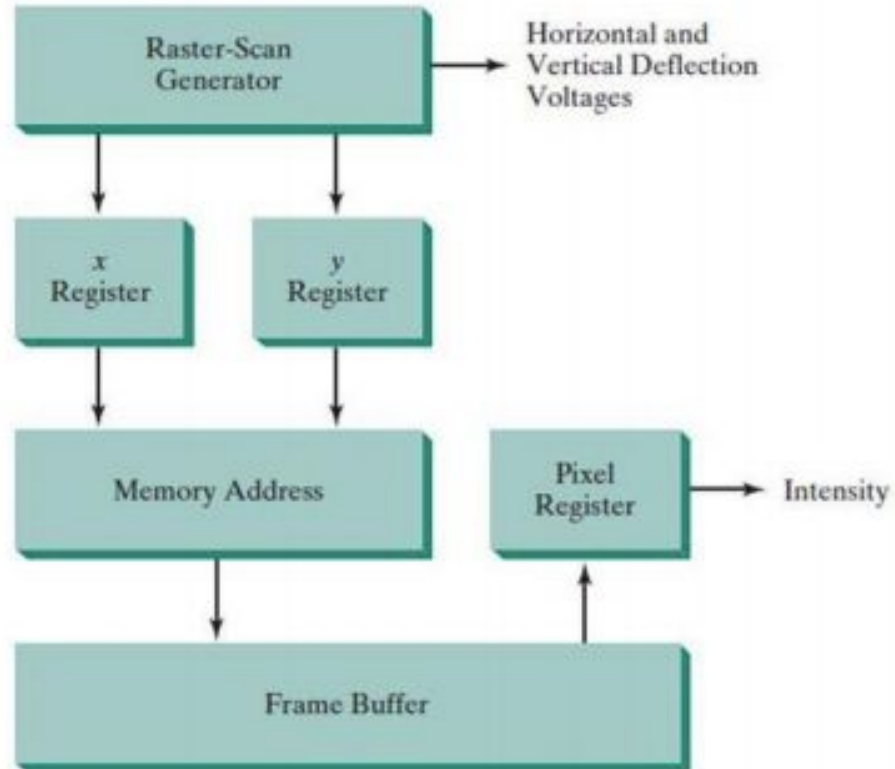
- Realistic images with different shades cannot be drawn.
- Colour limitations.

Video controller

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- The basic operations of the video controller
- Two registers are used to store the coordinate values for the screen pixels
- Initially, the x register is set to 0 and the y register is set to the value for the top scan line.
- The contents of the frame buffer at this pixel position are then retrieved and used to set the intensity of the CRT beam.
- Then the x register is incremented by 1, and the process is repeated for the next pixel on the top scan line.
- This procedure continues for each pixel along the top scan line.

Video controller



Terminologies

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- Refresh buffer / frame buffer
- Horizontal or vertical retrace
- Aspect ratio
- Resolution
 - Intensity
- Refresh Rate
- Persistence

Thank you

Computer Science Department, NAST